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Reference : Stewart §3.1-3.4, 4.1, 4.3

Marginal Quantities.

In economics, the term *marginal* is used to denote the rate of change of a quantity with respect to its dependent variable. In other words, the *marginal* of a function is simply the *derivative* of a function! For example,

- **Marginal Cost:** Suppose that $C(x)$ is the total cost function for a manufacturer to produce x units of a commodity. Then the *marginal cost of production* is $C'(x)$ which can be used to estimate the extra cost in producing one more unit of the commodity, i.e.

$$\Delta C = C(x+1) - C(x) = \frac{C(x+1) - C(x)}{1} \approx C'(x).$$

- **Marginal Revenue, Marginal Profit:** Suppose that $R(x)$ is the revenue generated when x units of a commodity is produced and sold, and $\Pi(x)$ is the corresponding profit. Then, the *marginal revenue* $R'(x)$ is used to approximate the additional revenue in producing one more unit of the commodity $\Delta R = R(x+1) - R(x)$. Similarly the *marginal profit* $\Pi'(x)$ is used to approximate $\Delta \Pi = \Pi(x+1) - \Pi(x)$.
- **Marginal Utility:** Let $U(x)$ be the utility of consuming x units of a commodity. Then, we use *marginal utility* $U'(x)$ to estimate the additional utility of consuming one more unit of the commodity.

Question 1. Suppose the weekly cost to manufacture x sets of Pulsar 40-in television is given by

$$C(x) = 10^{-4}x^3 - 0.09x^2 + 400x + 12000.$$

- (a) Use the marginal cost to estimate the extra cost of producing the 11th Pulsar TV, $C(11) - C(10)$.

By definition of marginal cost

$$\Delta C = C(x+1) - C(x) \approx C'(x)$$

$$C'(x) = 3 \times 10^{-4}x^2 - 0.18x + 400$$

$$C(11) - C(10) \approx C'(10) = 3 \times 10^{-4} \times 10^2 - 1.8 + 400 = 398.5$$

- (b) When is the marginal cost decreasing and when is it increasing?

To determine whether the marginal cost $C'(x)$ increase or decrease, calculate the first derivative of $C'(x)$, i.e. $C''(x)$

$$C''(x) = 6 \times 10^{-4}x - 0.18 = 6 \times 10^{-4}(x - 300)$$

When $0 < x < 300$, $C''(x) < 0 \rightarrow C'(x)$ is decreasing

$x > 300$, $C''(x) > 0$, $C'(x)$ is increasing

- (c) Let x be the weekly demand for Pulsar TV and p be its unit price. It is known that $0 \leq x \leq 15000$ and $p(x) = -0.04x + 600$. Write down the profit function $\Pi(x) = x \cdot p(x) - C(x)$. How many TV sets should be produced weekly in order to maximize the profit?

$$C(x) = 10^{-4}x^3 - 0.09x^2 + 400x + 12000$$

$$p(x) = -0.04x + 600$$

$$\Pi(x) = x \cdot p(x) - C(x) = -10^{-4}x^3 + 0.05x^2 + 200x - 12000$$

To find the critical point of $\Pi(x)$ in $[0, 15000]$, calculate the first derivative of $\Pi(x)$ and find x such that $\Pi'(x) = 0$

$$\Pi'(x) = -3 \times 10^{-4}x^2 + 0.1x + 200 = 0 \Rightarrow x = 1000 \vee -\frac{2000}{3}$$

$$\Pi(1000) = 10^5 \times 0.3 = 30000$$

Last, check the boundary condition of $\Pi(x)$ in $[0, 15000]$. We could find out both $\Pi(0)$ and $\Pi(15000)$ are less than 0

Thus, we can obtain the maximum profit 30000 when producing 1000 TV sets weekly.

Point Elasticity of Demand.

In economics, various scales of measurement are used in counting production quantity (e.g. in thousands vs. in millions) or pricing (in different currencies). Therefore, the derivative $f'(p)$ is not ideal to describe the sensitivity of the demand $y = f(p)$ of a commodity with respect to its price p .

To eliminate the dependence of the scales, economists use the **point elasticity of demand** to describe the price sensitivity of a commodity, which is defined as the percentage change of quantity demanded divided by the percentage change in prices.

In the case when the units are divisible, the **point elasticity of demand** is given mathematically as

$$\varepsilon = \lim_{\Delta p \rightarrow 0} \frac{\frac{y(p+\Delta p) - y(p)}{y(p)}}{\frac{\Delta p}{p}} = \frac{p}{y} \cdot \frac{dy}{dp}$$

A commodity is said to be **elastic** if $\varepsilon < -1$ and to be **inelastic** if $-1 < \varepsilon < 0$.

Question 2. Consider a linear demand function $y(p) = a + bp$ for $0 \leq p \leq -\frac{a}{b}$ where a and $b < 0$ are constants. For what values of p is the point elasticity (i) between 0 and -1 , (ii) smaller than -1 ?

$$y(p) = a + bp, \quad 0 \leq p \leq -\frac{a}{b}$$

$$\epsilon = \frac{p}{y} \cdot \frac{dy}{dp} = \frac{p}{y} \cdot b = \frac{bp}{a + bp}$$

$$(i) \quad -1 < \epsilon < 0$$

$$\therefore a + bp > 0, \quad b < 0$$

$$-1 < \frac{bp}{a + bp} < 0 \Rightarrow 0 < p < -\frac{a}{2b}$$

$$(ii) \quad \epsilon < -1$$

$$\therefore a + bp > 0, \quad b < 0$$

$$\frac{bp}{a + bp} < -1 \Rightarrow p > \frac{a}{2b} \Rightarrow \frac{a}{2b} < p \leq -\frac{a}{b}$$

Question 3. Suppose that $y(p)$ is the demand function for a commodity where p is in TWD and y is in thousands.

(a) Write down the demand function $Y(P)$ in terms of the demand function y where P is in USD and Y is in millions. (Assume that 1 USD = 30 TWD.)

(b) Compute $\frac{dY}{dP}$ in terms of $\frac{dy}{dp}$. Compare the point elasticity of $Y(P)$ with the point elasticity of $y(p)$.

$$(a) \quad 30P = p, \quad 1000Y = y \Rightarrow Y(P) = \frac{y(30P)}{1000}$$

$$(b) \quad \frac{dY(P)}{dP} = \frac{1}{1000} \cdot \frac{d(y(30P))}{dP} = \frac{1}{1000} \cdot y'(30P) \cdot \frac{d30P}{dP} = \frac{dY(P)}{dP}$$

$$\epsilon_P = \frac{P}{Y} \cdot \frac{dY(P)}{dP} \quad \text{For } 30P = p, \quad 1000Y = y$$

$$\epsilon_P = \frac{P}{30} \cdot \frac{1000}{y} \cdot \frac{dY(P)}{dP} = \frac{100}{3} \cdot \frac{P}{y} \cdot \frac{dY(P)}{dP} = 0$$

$$\text{Since } \frac{3}{100} \cdot \frac{dy(p)}{dp} = \frac{dY(P)}{dP}, \quad 0 \Rightarrow \frac{100}{3} \cdot \frac{P}{y} \cdot \frac{3}{100} \cdot \frac{dy(p)}{dp} = \frac{P}{y} \cdot \frac{dy(p)}{dp} = \epsilon_p$$

$$\Rightarrow \epsilon_P = \epsilon_p$$

Question 4.

- (a) Consider a demand function $y(p) = Kp^{-r}$ where K and r are positive constants.

What is its point elasticity? Does it depend on the price p ?

$$\epsilon = \frac{dy}{dp} \cdot \frac{p}{y} = -r \cdot Kp^{-r-1} \cdot \frac{p}{y} = -r \cdot K \cdot p^{-r-1} \cdot p \cdot \frac{1}{K} \cdot p^r = -r$$

$\rightarrow \epsilon$ does not depend on price.

- (b) Show that if the point elasticity is a constant $-c$, then the demand function is $y(p) = Kp^{-c}$ for some constant $K > 0$. (Hint. Show that $y(p) \cdot p^c$ is a constant function.)

$$\text{Suppose } \epsilon = \frac{dy}{dp} \cdot \frac{p}{y} = -c \Rightarrow \frac{dy}{dp} = -c \cdot \frac{y}{p}$$

$$\text{Consider } f(p) = y(p) \cdot p^c$$

$$f'(p) = y'(p) \cdot p^c + y(p) \cdot c \cdot p^{c-1} = -cy \cdot p^{c-1} + cy \cdot p^{c-1} = 0$$

From $f'(p) = 0$, we can conclude that $f(p)$ is a constant function

$$f(p) = k, \text{ Thus, } y(p) = kp^{-c}$$

Question 5. Prove that if a commodity is **inelastic**, then an increase in price p leads to an increase in total revenue $y(p) \cdot p$. Analogously, show that if a commodity is **elastic**, then an increase in price p leads to a decrease in total revenue $y(p) \cdot p$.

$$\frac{d}{dp}(y(p) \cdot p) = y'(p) \cdot p + y(p) \cdot p' = y(p) \left[\frac{y'(p)}{y(p)} \cdot p + p' \right]$$

When a commodity is elastic, i.e. $\epsilon(p) < -1$

total revenue $y(p) \cdot p$ is decreasing in terms of p

When a commodity is inelastic, i.e. $-1 < \epsilon(p) < 0$

total revenue $y(p) \cdot p$ is increasing in terms of p .